



Context-aware driver risk prediction with telematics data

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ABSTRACT

Driving risk prediction is crucial for safety and risk mitigation. While traditional methods rely on demographic information for insurance pricing, they may not fully capture actual driving behavior. To address this, *telematics data* has gained popularity. This study focuses on using telematics data and *contextual information* (e.g., road type, daylight) to represent a driver's style through tensor representations. Drivers with similar behaviors are identified by clustering their representations, forming risk cohorts. Past at-fault traffic accidents and citations serve as partial risk labels. The relative magnitude of average records (per driver) for each cohort indicates their risk label, such as low or high risk, which can be transferred to drivers in a cohort. A classifier is then constructed using augmented risk labels and driving style representations to predict driving risk for new drivers. Real-world data from major US cities validates the effectiveness of this framework. The approach is practical for large-scale scenarios as the data can be obtained at scale. Its focus on driver-based risk prediction makes it applicable to industries like auto-insurance. Beyond personalized premiums, the framework empowers drivers to assess their driving behavior in various contexts, facilitating skill improvement over time.

1. Introduction

A recent trend in the auto insurance industry is to use *telematics data* (i.e., data collected by OBD-II devices or smartphone applications) together with *demographic information* (e.g., age, gender, and marital status) to determine driving risk and insurance policy rates. The important benefit of using telematics data is to better demonstrate *natural driving behavior* (Toledo et al., 2008; Ziakopoulos et al., 2022a), which cannot be achieved otherwise if only demographic information is considered. However, understanding how to utilize telematics data to the best effect remains a challenging problem. While one group of existing studies has employed telematics data by summarizing them as *coarse-grained* representations (Verbelen et al., 2018; Hu et al., 2018; Ayuso et al., 2019; Guillen et al., 2019), another group of studies sought to use such information in a more *fine-grained* form Guo and Fang (2013), Klauer et al. (2014), Zheng et al. (2014), Joubert et al. (2016), Zhang et al. (2017), Gao et al. (2018a), He et al. (2018), Wang et al. (2018a), Chen et al. (2019), with neither set of studies having made the case for which way leads to a best practice.

To appropriately characterize a driver, it is essential to consider contextual information alongside telemetry data. Without understanding the context, a driver might be misjudged as safe, even if their driving behavior is risky under specific conditions (e.g., rainy days). Contextual properties like location, time, traffic, and weather conditions signif-

icantly influence driving actions. However, several previous studies have not adequately integrated contextual information into their future risk predictions (Guo and Fang, 2013; Klauer et al., 2014; Zheng et al., 2014; Zhang et al., 2017; Gao et al., 2018a; Wang et al., 2018b,a; Chen et al., 2019; Reig Torra et al., 2023). Moreover, incorporating extensive telematics or contextual data in studies such as Masello et al. (2023) may present challenges concerning scalability and the widespread applicability of their proposed framework.

Before delving into telematics-based driving risk prediction, it is essential to confront a significant challenge: accurately *defining* driving risk before effectively measuring it. Numerous prior studies have sought to define driving risk in diverse ways. For some, it pertains to the likelihood of being involved in a crash, a severe traffic offense such as driving under the influence (DUI), or, more broadly, the act of filing an *at-fault* insurance claim (Peck and Kuan, 1983; Gebers and Peck, 1992; Rajalin, 1994; Gebers and Peck, 2003; Guo and Fang, 2013; Klauer et al., 2014; Zheng et al., 2014; Joubert et al., 2016; Zhang et al., 2017; He et al., 2018; Gao et al., 2018a; Wang et al., 2018a; Chen et al., 2019). Other recent studies have approached risk by observing dangerous actions taken by drivers, with these actions being specified by human experts, such as harsh braking (Dingus et al., 2006; Guo and Fang, 2013; Klauer et al., 2014; Zheng et al., 2014; Stipancic et al., 2018; Ziakopoulos et al., 2022b). Furthermore, some studies enlisted

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human experts to assign risk scores to drivers based on their driving history as derived from telematics data (Wang et al., 2018a). In contrast to categorically defining risk according to these terms, we propose a different approach to define risk as a *factor* of an individual's previous driving record, including at-fault accidents and traffic violations, which contributes to the overall understanding of driving risk but does not solely explain it. Such information is readily available for a large group of drivers and proves valuable in assessing driving behavior. However, it is important to recognize that past traffic records can only serve as *partial ground-truth labels*.¹ In essence, while these labels provide valuable insights, they may not fully encompass the complexity of driving risk, necessitating a comprehensive approach in our research.

This paper presents a novel driving risk prediction framework that leverages telematics data alongside contextual information. Our main objective is to enhance the representation of driving behavior by combining telematics and contextual data, thereby advancing beyond the approaches outlined in earlier studies (Joubert et al., 2016; Zhang et al., 2017; Gao et al., 2018a,b; He et al., 2018; Hu et al., 2018; Wang et al., 2018b,a). The focus is on effectively aggregating telematics data within different contexts to create one or more representations for each driver, reflecting their driving style or footprint. It is important to clarify that our aim is not to develop a unique style, as seen in Moosavi et al. (2021). Instead, we aim to create representations that can later be used to infer driving risk, distinguishing between low-risk and high-risk behaviors. Thus, it is reasonable to anticipate that different drivers could share similar representations. Another key objective of this study is to rely on weak or partial risk signals. This design choice ensures that the proposed framework can be easily scaled, particularly if it finds adoption within the auto insurance industry. It is important to acknowledge that our work does not contribute to trajectory-based risk prediction (TBRP), which is a compelling domain. However, TBRP faces two significant challenges. Firstly, it requires a considerable effort to obtain relevant risk labels on a large scale. Secondly, extensive data collection, such as video streams from drivers during their driving or additional complex sensory data (e.g., LiDAR), may be necessary to enable TBRP. These challenges inherently hinder the scalability of a TBRP approach.

Our risk prediction framework comprises three major components, outlined as follows. The first part involves deriving fine-grained representations of telematics data in "context". Contextual information is directly encoded while creating these representations, incorporating details such as road type (e.g., low-speed versus high-speed), road shape (e.g., turn segments), and period of day or daylight (i.e., day or night). The representation includes trajectory information, encompassing attributes like speed, acceleration, angular-speed, and change-of-angle between two consecutive observations. Additionally, the shape of a road is encoded in the representation. To extract this shape, a previously proposed rule-based algorithm is utilized, achieving over 97% accuracy in detecting smooth and sharp turns in a trajectory (MoosaviNejadDaryakenari, 2020). An example of representing driver behavior in context is a representation that encodes the "*distribution of speed over high-speed road segments during a particular time of the day*".

The second part of the approach involves refining the available partial (or weak) risk labels. This is achieved by comparing a driver's driving behavior with that of a set of "presumably" safe drivers (i.e., drivers with no past traffic records over a period of 10 years). The result is a new representation which we call *deviation view*. Subsequently, a clustering method is applied to these deviation views to identify major risk cohorts. Each cohort is then labeled based on the partial labels of its members, which are categorized into broader risk categories (e.g., low-risk, medium-risk, and high-risk). This cohort label is subsequently transferred to each driver.

¹ Traffic citation or at-fault accident records alone cannot accurately indicate whether someone is a risky driver or not. Therefore, they are considered partial or weak ground-truth labels.

Lastly, a risk cohort prediction classifier is constructed, utilizing deviation maps as input and drivers' cohort risk labels (low-risk, high-risk, etc.) as the target. Various well-known classification models have been tested to select the most suitable model(s), ranging from traditional approaches like the Gradient Boosting Classifier to deep models like Convolutional Neural Networks. Subsequently, the proposed approach undergoes testing on a large real-world dataset of telematics data collected from five major cities in the United States (Philadelphia and Pittsburgh in Pennsylvania, Memphis in Tennessee, Atlanta in Georgia, and Columbus in Ohio).² Our analysis and results reveal significant advantages in contextualizing telematics data for improved risk prediction outcomes and demonstrate the effectiveness of our risk label refinement process in identifying meaningful risk cohorts. Furthermore, it is observed that utilizing a deep neural network model leads to better risk prediction outcomes. This improvement seems to stem from how the input data (i.e., encoded driver behavior) is represented as tensors, enabling the extraction of more meaningful spatiotemporal patterns when employing a deep model like the convolutional neural network.

This paper introduces the following key contributions:

- A novel technique that contextualizes telematics data by integrating diverse sources of contextual information, including road type, road shape, and daylight information.
- A clustering-based approach to enhance weak driving risk labels before constructing a risk prediction model.
- An end-to-end driving risk prediction process tailored for applications like usage-based insurance and driver coaching.

2. Related work

Driving risk prediction has been a topic of much research over the past few decades. While earlier studies mostly employed demographic information (e.g., driver's age, gender, marital status, car's age, and car's value) (Peck and Kuan, 1983; Gebers and Peck, 1992; Rajalin, 1994; Hu et al., 1998; Lourens et al., 1999; Elliott et al., 2000; Gebers and Peck, 2003), the recent studies have mostly used telematics data in addition to demographic information for driving risk prediction (Guo and Fang, 2013; Klauer et al., 2014; Zheng et al., 2014; Moosavi et al., 2017; Wang et al., 2018b; Gao et al., 2018a; Wang et al., 2018a; Ayuso et al., 2019; Hu et al., 2018; Guillen et al., 2019; Pesantez-Narvaez et al., 2019; Denuit et al., 2019; Pérez-Marín et al., 2019; Eling and Kraft, 2020; Guillen et al., 2021; Maillart, 2021; Moosavi et al., 2021; Singh and Kathuria, 2021; So et al., 2021). In this section, an overview of related work is provided by categorizing them based on the type of their input data.

2.1. Risk prediction based on demographic data

A large body of research on driving risk prediction has mainly relied on demographic data. Peck and Kuan (1983) proposed a multi-regression model that used data on accidents and violations from the past three years, as well as demographic factors, to predict the number of future accidents for a large group of drivers in California. They found that prior citation frequency, territorial accident rate, and prior accident frequency were the top three important factors in predicting future accidents. In another study, Gebers and Peck (1992) examined the risk of older drivers through several experiments. They found that traffic conviction record for older drivers is a stronger predictive factor for future accidents in comparison to younger drivers. Lourens et al. (1999) studied the impact of several factors on the risk of involvement in accidents. Using single and multivariate regression analyses, they

² This dataset is provided by our industrial sponsor, a major insurance company in Columbus, Ohio.

found that annual mileage, previous traffic conviction records, age, age-citation,³ and gender-education to be significant predictor factors of risk. Separating experienced and non-experienced drivers in studying the impact of previous traffic convictions on future crashes or serious violations was examined by Elliott et al. (2000). They found that a serious offense in the previous year doubled the odds for a serious offense in the subsequent year, and a previous at-fault record increases the odds by 50% for new at-fault crashes. A common aspect of all of these studies is that they have relied on demographic data, previous traffic conviction records, or questionnaire surveys to describe drivers, which may not accurately represent the naturalistic behavior of a driver. Without telematics data, fair and accurate prediction of driving risk is not possible due to inherent biases in applying predictions for a group of drivers to individual drivers and inadequate information about the past history of drivers. Furthermore, traditional approaches are limited in providing precise case-by-case prediction of driving risk. As a result, most studies have focused on analyzing and predicting risks for groups of drivers (Gebbers and Peck, 1992; Rajalin, 1994; Hu et al., 1998; Lourens et al., 1999; Elliott et al., 2000).

2.2. Risk prediction based on telematics data

Using telematics data alone for driving risk prediction is a recent trend given notable attention in the past few years (Toledo et al., 2008). Joubert et al. (2016) proposed using accelerometer and speed data as telematics for driving risk prediction. They used a two-dimensional matrix that only included accelerometer data along X and Y axes. The matrix was used to calculate a distribution of values for different cells, and risk categories were defined based on these values. The authors also incorporated speed and road type data to create matrices representing acceleration-based driving behavior for various types of roads and speeds. However, the study had limitations in terms of the amount and type of telematics and contextual data used. Hu et al. (2018) used telematics data for predicting future crash risk by analyzing patterns such as “fast-acceleration”, “hard-braking”, and “speeding” in a specific context using regression-based models. They also performed map-matching to incorporate speed-limit and expected traffic speed data. However, they did not use finer-grained telematics data in their study. Wang et al. (2018b) proposed using a CNN model to predict driving risk from accelerometer data. They created a scatter map from the raw data and converted it into a 2D image. Then, they used a CNN-based auto-encoder to initialize their pipeline and train a second CNN network to predict risk as a binary label. In another study, Zhang et al. (2017) proposed a model to use telematics data to create state graphs, which can be used to detect abnormal driving behavior for individual drivers. They compared an individual’s state graph with an aggregate state graph created from data of a large group of drivers to identify any unusual behavior. The authors also introduced pre-defined labels to classify the type of abnormal behavior, such as rapid acceleration, sudden braking, and rapid swerving. Wang et al. (2018a) also used the state-graph representation to predict driving risk, but they combined it with an auto-encoder neural network model to derive a more compact representation. They used this compact representation to predict driving risk using a Support Vector Regression model. However, their method required manual labeling of drivers as risky or safe, which may not be a scalable solution.

Guo and Fang (2013) analyzed the 100-car naturalistic driving study (NDS) dataset to perform risk assessment for individual drivers based on crash and near-crash (CNC) and critical-incident events (CIE). Using negative binomial regression and K-Means clustering, they identified important risk factors such as age, personality, and CIE, and classified drivers into low, moderate, and high-risk clusters. Klauer et al. (2014) used the same dataset to study the impact of secondary tasks on the risk

of crash or near-crash events, and found involvement in tasks such as texting, eating, and using a cell phone increased the risk for both novice and experienced drivers. Chen et al. (2019) also used the NDS dataset and developed an auto-encoder model to optimize input data window-size and a deep-neural-network to predict CNC events for near-crash and crash event prediction.

2.3. Risk prediction based on demographic and telematics data

The last group of studies employed both telematics and demographic information for driving risk prediction. Gao et al. (2018a) used distribution matrices of speed and acceleration values to propose a framework that predicts the number of insurance claims by employing a Poisson General Additive Model (GAM). They condensed the telematics data representation and used driver and vehicle demographics as input parameters. In a follow-up study, Gao et al. (2018b) improved on their framework by extending demographic factors and representing telematics data with a wider range of speed values. They used a GAM model to predict the frequency of claims and concluded that lower speed ranges offered better predictivity for driving risk. Ayuso et al. (2019) used telematics and demographic data to predict insurance claim frequency. Coarse-grained telematics factors included annual distance driven, distance driven at night, distance over the speed limit, and urban distance. They employed a Poisson regression model and found age, vehicle type, distance driven over the speed limit and urban distance to be the most significant factors. Guillen et al. (2019) used a Zero Inflated Poisson regression model to correct for biases and avoid risk prediction based solely on the number of miles driven. Their coarse-grained telematics factors included distance driven over the speed limit, urban distance, and distance driven per year. Verbelen et al. (2018) used regression-based models with coarse-grained telematics factors such as yearly distance, number of trips, and coverage of different road types. He et al. (2018) proposed a comprehensive framework for risk prediction that extracted fine-grained information from trajectories using driving state variables and a machine learning model. However, a limitation of all these studies was the limited usage of contextual information.

2.4. Related work summary

Our risk prediction framework falls into the category of using only telematics data to predict driving risk. However, our approach differs from existing solutions in several important ways. Firstly, we use finer-grained representations of telematics data to better capture driving behavior. Secondly, we incorporate contextual data to more accurately represent and evaluate driving behavior in context. Finally, we address the issue of weak risk labels by augmenting them before modeling and prediction. This is often overlooked when using traffic records as ground-truth labels to train or test risk prediction models. We recognize that traffic records or their absence alone are insufficient indicators of risk, and correcting this bias is necessary to build accurate risk prediction models.

3. Research problem

Let us suppose that telematics data is composed of trajectories. A trajectory T is a time-ordered sequence of data points $\langle p_1, p_2, \dots, p_{|T|} \rangle$. Each data point p is represented by a tuple $(t, Speed, Heading, Lat, Lng)$, where t is a timestamp, $Speed$ is the ground velocity of a vehicle (in km/h), $Heading$ is the heading (or bearing) of a vehicle (a value between 0 and 359 degrees, and Lat and Lng represent latitude and longitude. Further, any data that can be used in conjunction with telematics data to better demonstrate environmental factors are called “contextual information” in this paper. Examples are traffic flow, daylight, and road characteristics (e.g., curvature). The risk prediction problem is formulated as follows:

³ The driver’s age and citation history taken together as a single factor.

Given:

- A set of drivers D , with
- The *observed risk score* for each driver $D \in D$ denoted by R_D . Here, R_D represents the total number of at-fault traffic accidents and violation records for driver D in the past five years.
- A set of trajectories $\mathcal{T}_D$ for each driver $D \in D$.
- Contextual information C that could be used in conjunction with the trajectory data.

Build:

- A contextualized telematics representation C_D for driver $D \in D$ based on $\mathcal{T}_D$ and accompanying context information C (e.g., road shape, road type, and time information).
- A classifier $\mathcal{C}$ to predict a risk cohort label $\mathcal{L}_D$ for driver $D \in D$, using the representation above C_D . Here, risk cohorts are categorical labels (such as *low-risk* and *high-risk*).

Objective:

- Minimize the average risk score for the low-risk cohort and maximize it for the high-risk cohort, while maintaining a size balance among cohorts.

The extension of this objective to more than two risk cohorts is straightforward. Additionally, the requirement for balanced driver cohorts ensures that there are no scenarios where only a few safe (or risky) drivers end up in one cohort, while the rest are in the other. Furthermore, the framework is validated by utilizing statistical significance testing to confirm the significant difference in risk scores between the low-risk and high-risk cohorts.

4. Methodology

This section describes our context-aware driving risk prediction framework that consists of three components. The first component creates a representation for telematics data in context. The second component builds a risk cohort classifier based on contextualized telematics information and weak (or partial) risk labels. Finally, the third component performs risk cohort prediction for unseen drivers in real-time as a real-world application. Fig. 1 represents the big-picture of our approach, where the first two components make up the *Modeling Process* and the third component makes up the *Prediction Process*.

4.1. Contextualized telematics representation

This section describes how to use telematics and contextual data to build representations for drivers. The telematics data consists of trajectory datasets, and the representations are built based on different contextual conditions (e.g., road type and road shape). Next, the trajectory datasets are described, followed by the description of how to build representations for each trajectory. Lastly, the process of building representations for each driver is explained.

4.1.1. Modeling trajectory dataset

The modeling dataset consists of a large set of drivers (e.g., 4,000 drivers) with a minimum number of trajectories (e.g., 100 trajectories) per driver. The drivers in this dataset must satisfy the following constraints: (1) each driver must own exactly one vehicle, and (2) the driver should not share their vehicle with any other driver. These constraints help ensure that trajectories collected for a vehicle had been generated by a single driver, who only drove that vehicle, thereby reducing the chance of vehicle-related biases.⁴ This set is also referred to as the *observation set* in this paper.

4.1.2. Reference trajectory dataset

The reference trajectory dataset consists of a large set of *presumably safe* drivers (e.g., 5,000 drivers), also with a minimum number of trajectories for each driver. Drivers in this set must have no record of at-fault accidents or traffic violations in the past 10 years, hence may be considered safe.

4.1.3. Raw feature map: Representing a trajectory

The initial step in utilizing telematics data involves designing a trajectory representation method. Inspired by the work of Joubert et al. (2016), we adopt a *feature map representation* for trajectories. This representation employs a matrix, where each axis corresponds to a specific attribute (e.g., speed, acceleration, or angular speed), and each cell indicates the *normalized frequency* of trajectory data points falling into that cell. For instance, if the feature map's x- and y-axes represent speed and acceleration, respectively, a cell would encompass specific speed and acceleration intervals (e.g., speed interval = [5, 25] km/h, and acceleration interval = [0.0, 0.2] m/s²). The cell value denotes the normalized frequency of trajectory data points falling within these intervals (with raw frequencies normalized by the maximum frequency value). Fig. 2-a demonstrates an example of a raw feature map.

4.1.4. Deviation feature map: Representing a driver

While raw feature maps can effectively represent trajectories, our preference lies in devising a means to represent a “driver” using telematics data, as risk labels are generally unavailable on a trajectory-basis in our study, and many real-world scenarios share this limitation. Moreover, instead of a simple aggregation over raw feature maps, we seek to incorporate more information into this macroscopic representation, akin to capturing a driver's (relative) signature or style. Given a set of raw feature maps generated for a collection of trajectories from driver d (sampled from the observation set), we propose a novel process for creating a new driver representation called a *deviation feature map*. The key steps of this process are outlined in Algorithm 1. To execute this process, the reference set is randomly divided into *base* and *control* sets; the former containing trajectories from 70% of the presumably safe drivers, and the latter comprising trajectories from the remaining drivers in the reference set. The process starts by generating raw feature maps for trajectories in the base set (lines 1–7). Then, it generates a *histogram matrix* that shows the distribution of raw feature maps in the base set (line 9). Example of a histogram matrix is shown in Fig. 3. The next step is to generate raw feature maps, and then a histogram matrix for each driver in the control set (lines 11–20). Then, it obtains the difference between the histogram created for each driver in the control set, and the histogram created for the whole base set (lines 22–26), and summarize these differences to obtain a single matrix termed as the *natural deviation* (line 27). Next, the same process is repeated for the observation set by creating raw feature maps (lines 29–33), and then generating a histogram matrix for driver d (line 34), finally obtaining the difference between the matrix generated for driver d and the base histogram matrix (line 35). The output of this algorithm is the difference between the natural deviation and the observed deviation, termed as the *deviation feature map* for driver d (line 36).

The functions used in Algorithm 1 are described below.

- Function *build_raw_feature_map()*: This function creates a raw feature map for an input trajectory.
- Function *obtain_histogram_matrix()*: Given a set of raw feature maps, we summarize them by obtaining a histogram that shows the distribution of values in each cell.
- Function *obtain_difference()*: This function obtains deviation from the base histogram matrix. As the goal is to obtain difference between two distributions, any reasonable method can be employed to obtain such deviation. Examples of possible solutions are *Kullback–Leibler divergence* (Kullback and Leibler, 1951), *total variation* (Rowland, 2023), and *Hellinger distance* (Hellinger, 1909).

⁴ A driver might exhibit different behavior when driving different cars.

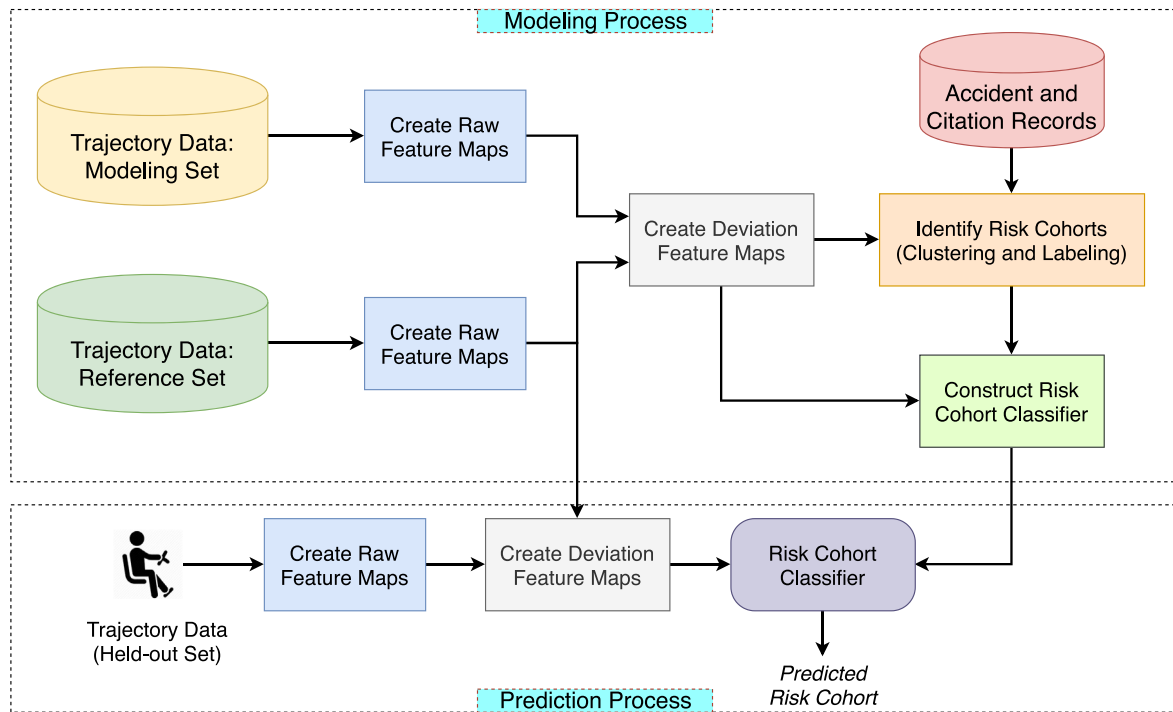


Fig. 1. The proposed risk prediction framework. The “Modeling Process” builds contextualized telematics representation for drivers and constructs a risk-cohort classifier; and the “Prediction Process” predicts risk cohort for new or unseen drivers.

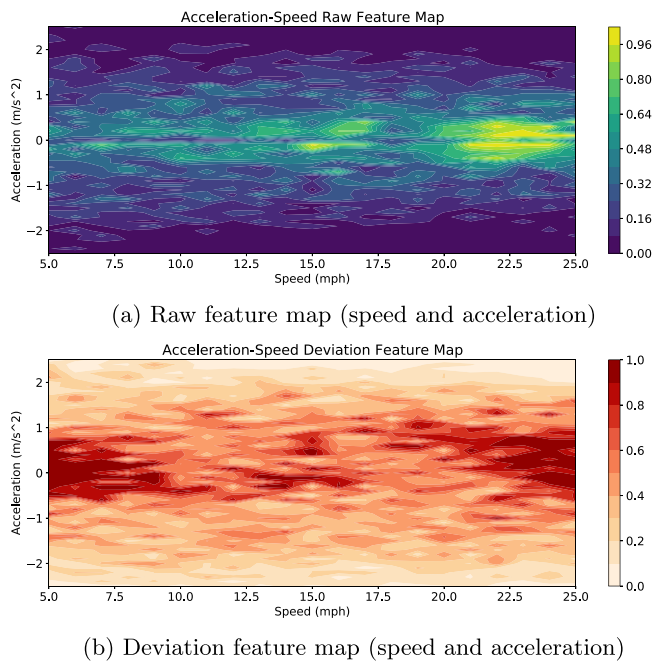


Fig. 2. Example of a “raw” and a “deviation” feature map for a trajectory and a driver, respectively.

- Function *summarize()*: This function summarizes deviation matrices of drivers in the control set, using operations such as summation, average, etc.
- Function *difference()*: This function measures the difference between the natural deviation (i.e., output of the *summarize* function), and the observed deviation for driver *d*, to build a deviation feature map for driver *d*.

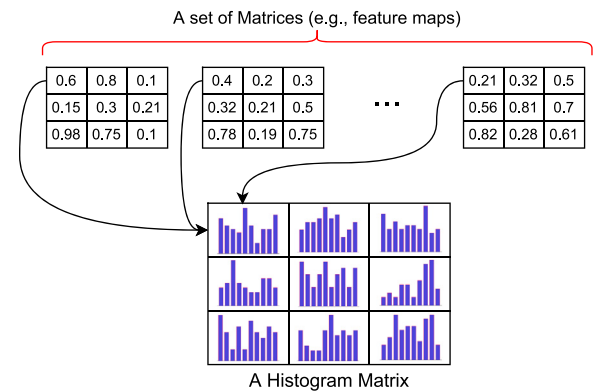


Fig. 3. Example of a histogram matrix, created based on a set of input matrices. Each cell in histogram matrix represents data distribution across corresponding cells in input matrices.

One example of deviation feature map that is generated for a driver in the observation set is shown in Fig. 2-b. The number of deviation feature maps to represent a driver depends on the number of raw feature maps to represent each trajectory. Further details on various types of raw feature maps used in this work are provided in Section 5.2.

4.2. Risk cohort classifier

As shown in Fig. 1, the main outcome of the modeling process is a risk cohort prediction classifier. The different requirements and steps of building such a classifier are described in the following sub-sections.

4.2.1. At-fault accident and citation records

This set contains traffic accident and citation records for each driver in the modeling set. In terms of accidents, only the *at-fault* cases are considered. For each driver, only their records in the last five years are taken into account.

Algorithm 1: Generating Deviation Feature Map

Input: The base set B , the control set C , and the observation trajectory set T_d

```

1  $B_{FM} = []$   $\triangleright$  Set of raw feature maps for the Base set  $B$ 
2 for driver  $k$  in  $B$  do
3   for trajectory  $t$  in  $T_k$  do
4      $fm = \text{build\_raw\_feature\_map}(t)$ 
5      $B_{FM}.\text{append}(fm)$ 
6   end
7 end
8  $\triangleright$  Summarize raw feature maps by creating a histogram matrix for the base set
9  $\text{hist\_matrix}_B = \text{obtain\_histogram\_matrix}(B_{FM})$ 
10  $\triangleright$  For each driver in control set, obtain raw features maps and their histogram matrix
11  $C_{HM} = []$   $\triangleright$  Set of histogram matrices for the Control set  $C$ 
12 for driver  $k \in C$  do
13    $k_{FM} = []$ 
14   for trajectory  $t \in T_k$  do
15      $fm = \text{build\_raw\_feature\_map}(t)$ 
16      $k_{FM}.\text{append}(fm)$ 
17   end
18    $\text{hist\_matrix}_k = \text{obtain\_histogram\_matrix}(k_{FM})$ 
19    $C_{HM}.\text{append}(\text{hist\_matrix}_k)$ 
20 end
21  $\triangleright$  Obtain the natural deviation by summarizing deviation of the control set from the base set
22  $C_{diff} = []$ 
23 for hist\_matrix  $hm \in C_{HM}$  do
24    $diff = \text{obtain\_difference}(\text{hist\_matrix}_B, hm)$ 
25    $C_{diff}.\text{append}(diff)$ 
26 end
27  $\text{natural\_deviation} = \text{summarize}(C_{diff})$ 
28  $\triangleright$  Generate the deviation feature map for driver  $d$ , based on the natural deviation
29  $d_{FM} = []$   $\triangleright$  Set of raw feature maps for the observation trajectory set
30 for trajectory  $t \in T_d$  do
31    $fm = \text{build\_raw\_feature\_map}(t)$ 
32    $d_{FM}.\text{append}(fm)$ 
33 end
34  $\text{hist\_matrix}_d = \text{obtain\_histogram\_matrix}(d_{FM})$ 
35  $\text{observed\_deviation}_d = \text{obtain\_difference}(\text{hist\_matrix}_B, \text{hist\_matrix}_d)$ 
36  $\text{deviation\_feature\_map}_d = \text{difference}(\text{observed\_deviation}_d, \text{natural\_deviation})$ 
Output:  $\text{deviation\_feature\_map}_d$ 

```

4.2.2. Identifying risk cohorts

Algorithm 2 is employed to identify risk cohorts for drivers in the modeling set. Each driver is represented by a set of D deviation maps. The deviation maps are used to cluster the drivers (line 3). After each clustering, a post-processing step is performed to associate each cluster with a risk cohort label (line 4). To do this, the partial (or weak) ground truth risk label data (i.e., traffic conviction records) is employed. The average number of records in each cluster is obtained, and then each cluster is associated with a categorical label (e.g., low-risk and high-risk).⁵ For each driver, their label set is updated by adding the newly

⁵ For instance, if we define our labels as low-risk and high-risk, then the cluster with a smaller average of conviction records per driver will be labeled as low-risk, and the other cluster will be labeled as high-risk.

obtained cohort label (lines 5–7). Finally, the most frequent label for each driver is found given a confidence coefficient C (lines 9–16) as a threshold. Based on this threshold, drivers who have been confidently assigned to a specific risk cohort are identified. For example, if a driver is assigned to the *low-risk* cohort (say) 80% of the time, they are labeled as a *low-risk* driver. If a driver is not confidently assigned to any of the cohorts, then their final label would be *Null*. The output of this algorithm is a subset of drivers in the modeling set with a valid final risk cohort label.

Algorithm 2: Identifying Risk Cohorts

Input: Modeling set drivers $\mathcal{M}$, Deviation feature maps $\mathcal{D}$, number of cohorts k , traffic conviction records $Recs$, and labeling confidence C .

```

1  $\text{driver\_to\_cohort\_label} = \{\}$ 
2 for deviation\_map  $D \in \mathcal{D}$  do
3    $res = \text{do\_clustering}(\mathcal{M}_D, k)$   $\triangleright \mathcal{M}_D$  represents the modeling set drivers by deviation map  $D$ 
4    $\text{cohort\_label} = \text{clustering\_label\_to\_risk\_cohort\_label}(res, Recs)$ 
5   for driver  $d \in \mathcal{M}$  do
6      $\text{driver\_to\_cohort\_label}[d].\text{append}(\text{cohort\_label}[d])$ 
7   end
8 end
9  $\text{final\_labeled\_set} = \{\}$ 
10 for driver  $d \in \mathcal{M}$  do
11    $labels = \text{driver\_to\_cohort\_label}[d]$   $\triangleright$  Labels assigned to driver  $d$  by clustering tasks
12    $\text{final\_label} = \text{find\_frequent\_label}(labels, C)$ 
13   if  $\text{final\_label} \neq \text{Null}$  then
14      $\text{final\_labeled\_set}[d] = \text{final\_label}$ 
15   end
16 end
Output:  $\text{final\_labeled\_set}$ 

```

While any clustering algorithm may be used in the *do_clustering()* function, *K-means* is a reasonable choice since the number of clusters must be specified beforehand. Regarding the number of clusters, any value greater than 1 can be chosen, depending on the driving context and application requirements. In this work, a value of 2 is used to test the framework.

4.2.3. Constructing the risk cohort classifier

The last step of the modeling process is to build a classifier that uses deviation maps as input to predict the risk cohort label (or the refined risk label). The input of this part is the set of labeled drivers and their deviation maps obtained from the previous step. Any off-the-shelf classifier can potentially be used to predict the risk cohort.

Emphasizing the significance of *refining risk labels* in our framework is essential. To illustrate its importance, consider a classifier built to predict driving risk using telematics data as input and past traffic records as risk labels. Although this approach seems straightforward, it may not yield satisfactory outcomes due to the *unreliability* of raw risk labels. An important practical observation is that having records of citations or at-fault accidents does not necessarily indicate a risky driver, and the absence of such records does not necessarily mean a driver is safe. As a result, a refining process for the labels is necessary before constructing a risk cohort classifier.

4.3. The prediction process

In order to employ the proposed framework for the real-world applications, we describe a *prediction process*. Given a set of trajectories for a new or an unseen driver, the important steps of this process are (1) Creating raw feature maps, (2) Creating deviation feature maps, and (3) Risk prediction using risk cohort classifier.

Table 1

The telemetry data that is employed in this study to develop and assess the effectiveness of the proposed framework. The dataset was gathered from five cities across the United States between July 2017 and March 2019. For data collection, dedicated devices were installed on each (personal) vehicle, encompassing a range of sensors, including GPS and accelerometer, in addition to capturing CAN-bus data through the OBD-II port.

# Vehicles	# Trajectories	Total driving distance (km)	Total driving hours
8,970	891K	10.7M	264K

Table 2

A snapshot of the telemetry data utilized in this research. The data was collected at a rate of 1 Hz, resulting in one record per second. Examples of attributes gathered include timestamp, GPS latitude (lat) and longitude (lng), GPS-based heading (or bearing, head), engine rounds per minute (RPM), vehicle speed (in km per hour), and acceleration (in meters per second square).

Index	Time	Lat	Lng	Head	RPM	Speed	Accel
1	15:14:43	40.50177	-80.07582	120	783	0	0
2	15:14:44	40.50177	-80.07582	120	1598	2	0
3	15:14:45	40.50177	-80.07582	120	1675	8	1.67
4	15:14:46	40.50176	-80.07581	114	1928	13	1.39
5	15:14:47	40.50175	-80.07579	108	1774	18	1.39
6	15:14:48	40.50174	-80.07575	108	1703	22	1.11
7	15:14:49	40.50172	-80.07568	108	1637	25	0.83
8	15:14:50	40.50169	-80.07562	108	1779	29	1.11
9	15:14:51	40.50167	-80.07554	108	1910	32	0.83
10	15:14:52	40.50164	-80.07545	114	1720	36	1.11

Table 3

Summary of duration (in minutes) and distance (in kilometers) statistics from the telemetry dataset employed in this study. The columns starting with “P” represent percentiles, where, for example, P25 indicates the 25th percentile.

	Mean	P25	P50	P75	P90
Duration (minutes)	16.4	6	12.5	23	35
Distance (kilo-meters)	12	1.6	6	14	28

5. Datasets and data representation

In this section, the datasets are described first, followed by the description of telematics data representation via feature maps.

5.1. Datasets

We utilized multiple comprehensive datasets to construct and assess the effectiveness of our risk prediction framework. These datasets were collected from various sources, including the *CAN-bus*,⁶ as well as other sensors such as GPS, accelerometer, and magnetometer. The data was collected at a rate of 1 Hz (i.e., one record per second), and a snapshot of sample data is presented in Table 2. The data was collected from five different cities: Atlanta (Georgia), Columbus (Ohio), Memphis (Tennessee), Philadelphia (Pennsylvania), and Pittsburgh (Pennsylvania). Table 1 provides a detailed summary of the telemetry dataset. In total, this extensive dataset comprises approximately 264,000 hours of driving, spanning over 10.7 million kilometers. Table 3 offers detailed statistics on duration and distance for each trajectory in the dataset.

In addition to telemetry data, we had access to demographic information on drivers, as well as their historical records of traffic crashes and citations, specifically (at-fault) incidents. To build the framework, the data was categorized into distinct sets, outlined as follows:

- **Modeling Set:** This set is utilized during the modeling process to construct the risk cohort prediction classifier. It comprises approximately 3750 drivers, each owning exactly one vehicle present in our dataset. None of the drivers in this set share their vehicle with

others. For every driver, precisely 100 distinct trajectories were collected.

- **Reference Set:** This set comprises drivers with a clean history, also referred to as “presumably safe” drivers, as they have no records of at-fault crashes or traffic citations between 2010 and 2019. Consisting of 5000 vehicles, some of which may be shared among multiple drivers, the constraint is upheld that all drivers sharing the same vehicle satisfy the aforementioned criteria. For analysis, each vehicle in this set contributes 100 distinct trajectories. Note that the number of drivers in this set also totals approximately 5000. The vehicle-sharing constraint in this set was intentionally relaxed to build a large collection of vehicles and drivers, as we empirically learned it leads to improved risk prediction outcomes.
- **Held-out Set:** This smaller set is reserved for evaluating the prediction process. It encompasses 250 drivers, each of whom owns exactly one vehicle, and no vehicle is shared among drivers in this set.

To establish partial ground-truth labels for drivers in both the modeling and held-out sets, their traffic citation and at-fault accident records from 2014 to 2019 were gathered. Remarkably, approximately 70% of the drivers in both sets were found to have at least one such record within the five-year time window.⁷

5.2. Data representation by raw feature maps

In this study, a trajectory is represented using 22 different feature maps, which are categorized into six groups as described in Table 4. A few notes regarding the definition of raw feature maps are as follows:

- **Attributes’ Order:** The order of attributes in each feature map suggests that the first attributes is used in *x*-axis and the second attribute in *y*-axis (see Section 4.1.3).
- **Context:** Shows the context that is used to generate a feature map (if any). Based on this, feature maps in group G1 (see Table 4) are context-agnostic.
- **Low-speed Roads:** A road segment is labeled as a *low-speed road* if its speed-limit falls in the interval [0, 50] mph. Speed-limit data was obtained using the Graphhopper Map-matching Library (Anon, 2023). Each trajectory was map-matched using location and time data, and speed-limit information was obtained for each segment that constituted a trajectory.
- **High-speed Roads:** A road segment is labeled as a *high-speed road* if its speed-limit falls in the interval [50, 90] mph. The definition of road type in this work is a practical choice after trying a few settings and analyzing their impact on final risk prediction results.⁸
- **Day Time:** The *civil twilight system*⁹ was used to annotate each trajectory based on its start time. In this sense, “day time” indicates a trajectory was generated during the day time based on the civil twilight system. Daylight information for each trajectory was obtained from Time And Date website (2023) for annotation.
- **Night Time:** This contextual concept indicates that a trajectory was generated during the night time based on the civil twilight system.
- **Turn Segments:** This contextual concept refers to representing trajectory attributes for turn segments that are extracted from a trajectory. A fairly accurate turn detection algorithm proposed in MoosaviNejadDaryakenari (2020) (see Section 6.4.4) is used to extract turn segments.

⁷ Please note that we cannot provide additional details about the drivers, including their demographics and driving history, due to confidentiality and privacy constraints.

⁸ Interested readers may try other settings to study the impacts.

⁹ See more at https://en.wikipedia.org/wiki/Twilight#Civil_twilight.

⁶ The Controller Area Network (CAN-bus) is a communication mechanism that transfers a variety of information related to a vehicle’s operation.

Table 4
Description of raw feature maps (attributes and contexts).

Group	Feature map	Attributes	Context	1st attr range	2nd attr range
G1	T1	Speed, Acceleration	N/A	[5, 25] mph	[−2.5, 2.5] m/s ²
	T2	Speed, Acceleration	N/A	[25, 45] mph	[−2, 2] m/s ²
	T3	Speed, Acceleration	N/A	[45, 80] mph	[−1, 1] m/s ²
G2	T4	Speed, Acceleration	Low-speed roads	[5, 25] mph	[−2.5, 2.5] m/s ²
	T5	Speed, Acceleration	Low-speed roads	[25, 45] mph	[−2, 2] m/s ²
	T6	Speed, Acceleration	High-speed roads	[5, 25] mph	[−2.5, 2.5] m/s ²
	T7	Speed, Acceleration	High-speed roads	[25, 45] mph	[−2, 2] m/s ²
	T8	Speed, Acceleration	High-speed roads	[45, 80] mph	[−1, 1] m/s ²
G3	T9	Speed, Acceleration	Day time	[5, 25] mph	[−2.5, 2.5] m/s ²
	T10	Speed, Acceleration	Day time	[25, 45] mph	[−2, 2] m/s ²
	T11	Speed, Acceleration	Day time	[45, 80] mph	[−1, 1] m/s ²
	T12	Speed, Acceleration	Night time	[5, 25] mph	[−2.5, 2.5] m/s ²
	T13	Speed, Acceleration	Night time	[25, 45] mph	[−2, 2] m/s ²
	T14	Speed, Acceleration	Night time	[45, 80] mph	[−1, 1] m/s ²
G4	T15	Speed, Acceleration	Turn segments	[5, 25] mph	[−3, 3] m/s ²
	T16	Speed, Acceleration	Turn segments	[25, 45] mph	[−3.5, 3.5] m/s ²
	T17	Speed, Acceleration	Turn segments	[45, 80] mph	[−2, 2] m/s ²
G5	T18	Speed, AngleChange	Turn segments	[5, 25] mph	Up to 20 degrees
	T19	Speed, AngleChange	Turn segments	[25, 45] mph	Up to 6 degrees
	T20	Speed, AngleChange	Turn segments	[45, 80] mph	Up to 4 degrees
G6	T21	Acceleration, AngleChange	Turn segments	[−2, 2] m/s ²	Up to 7 degrees
	T22	AngularSpeed, AngleChange	Turn segments	[0, 0.025] rph	Up to 15 degrees

- **AngleChange:** This basic feature is extracted from a trajectory and shows the change of *angle* between two consecutive observations. Angle is obtained based on three location points (e.g., A, B, and C) and using Eq. (1). Thus, the angle represents the orientation of a moving vehicle, and a change in angle implies a change in orientation.

$$\theta = \arccos\left(\frac{\vec{AB} \cdot \vec{BC}}{\|\vec{AB}\| \|\vec{BC}\|}\right) \quad (1)$$

- **AngularSpeed:** Another basic feature measures angular displacement per unit of time. The angular speed in radian per hour (rph) was calculated using Eq. (2). Here, $H(\cdot)$ calculates the haversine distance in terms of radians between two coordinates P_1 and P_2 , and r represents the Earth's radius (i.e., 6.3781×10^6 meters).

$$Angular_Speed(P_1, P_2) = H(P_1, P_2) \times \frac{3600}{r} \quad (2)$$

6. Experiments and results

In this section, the proposed risk prediction framework is evaluated based on real-world data. Some preliminaries and setups are discussed first, followed by risk prediction results on the held-out set. Lastly, a couple of scenarios are discussed to demonstrate how the proposed framework can be utilized in practice.

6.1. Preliminaries and setups

After representing a trajectory using the 22 different raw feature maps, the risk cohort classifier was created from the modeling and reference sets (see Section 4 for details). Different combinations of raw feature maps, and the set of all feature maps to represent trajectories were examined. Next, several settings are described before presenting the results.

- **Choice of Deviation Metric:** To obtain the deviation between two probability distributions in Algorithm 1, any deviation calculation method can be employed. However, for the sake of simplicity, *Hellinger distance* (Hellinger, 1909) is used for discrete distributions (Eq. (3)):

$$Hel_Dist(A, B) = \frac{1}{2} \sqrt{\sum_{i=1}^N (\sqrt{A[i]} - \sqrt{B[i]})^2} \quad (3)$$

Here A and B are two vectors.

- **Labeling Confidence C:** The labeling confidence in Algorithm 2 determines the final labeled set of drivers. This value is determined with respect to the number of deviation maps used to represent each driver (which depends on the number of raw feature maps to represent each trajectory) and the number of risk cohorts.

- **Number of Cohorts:** One of the advantages of this framework is the flexibility to define any number of cohorts. In this work, two cohorts were used, referred to as *low-risk* and *high-risk*.

- **Clustering Algorithm:** The choice of the clustering algorithm to be employed by Algorithm 2 is another customizable aspect of this framework. While K-Means clustering is used in this work, any other method that allows the user to set the number of clusters could be suitable.

- **Classification Method:** We used two different models for classification, Gradient Boosting Classifier (GBC) and Convolutional Neural Network (CNN).

- GBC: Gradient boosting is a strong general-purpose classifier, which usually performs very well on binary or multi-label classification tasks. The input to GBC should be in the form of a vector, which is created by converting deviation feature maps to a vector. Each vector represents one driver, and the risk label for each driver is obtained from the clustering (see Section 4). Grid-search was used to obtain the optimal parameters of the GBC classifier using a 5-fold cross-validation over the modeling set.¹⁰

- CNN: Given that the input data is in the form of matrix representation, where each matrix represents one feature map, the use of a convolutional neural network (CNN) seems to be a reasonable choice for this problem. Apart from the type and representation of input data, the use of CNN may further help to extract latent information from the input and use it to better classify drivers. After hyper-parameter tuning via grid search over the modeling set, the final CNN model includes two convolutional layers (each with 16 filters, kernel size 5×5 , and stride=1) with max pooling on top of convolutional output (each with pool size 2×2 and stride = 2), followed by three dense layers. All layers use *ReLU* as the activation function,

¹⁰ Optimal parameters were found as: *learning_rate* = 0.1, *num_estimators* = 300, and *max_depth* = 5.

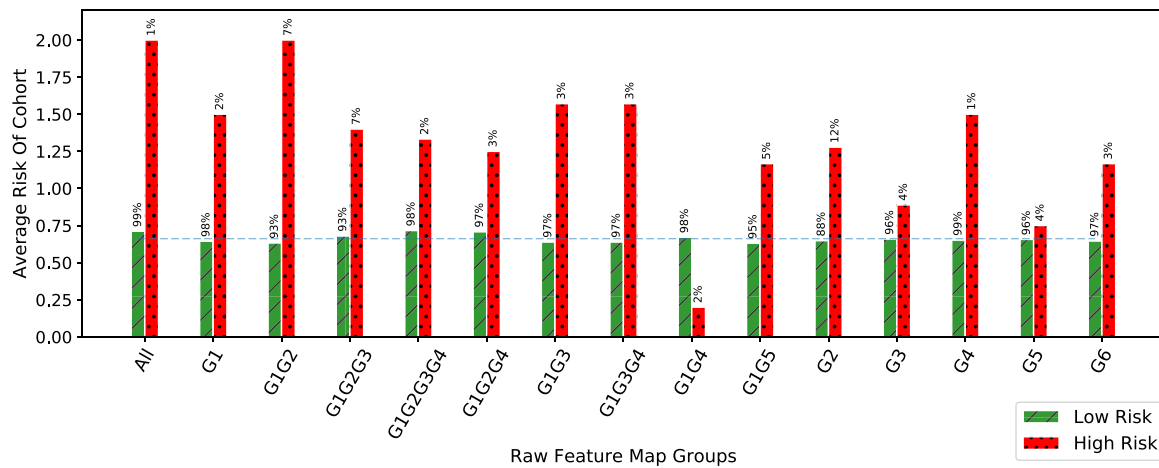


Fig. 4. Baseline risk cohort prediction results for drivers in the held-out set. X-axis shows the combination of raw feature maps, and Y-axis shows the average number of traffic records in each cohort (i.e., risk score – average number of traffic citations and at-fault accidents), and blue dashed lines show the average number of records in the held-out set (% numbers on bars show relative size of each cohort – i.e., what percentage of drivers in the held-out set were categorized into a specific risk cohort).

except the last dense layer which uses *softmax*. Lastly, Root Mean Square Propagation optimizer (Tieleman and Hinton, 2012) (RMSProp) with an initial learning rate of 0.001 is used as the optimization method in training.

- **Baseline Method:** As a baseline, a variation of the proposed solution is chosen, which employs raw feature maps and uses the original risk labels to build the risk classifier. In other words, the goal is to analyze the impact of two components through this approach: (1) transforming raw feature maps to deviation maps, and (2) refining risk labels by clustering. Based on this approach, each driver is represented by an average feature map for each type listed in Table 4. Additionally, a binary label based on drivers' past records is used to label them, where a driver with no record would be labeled as 0 (i.e., safe) and 1 otherwise.
- **Risk Score:** In this study, each driver is assigned a non-zero integer value representing the number of their at-fault traffic accidents and citations received over a five-year period, which includes the data collection period. The minimum value, by definition, is zero, while the maximum can be any positive integer.

6.2. Risk prediction results

Different combinations of raw feature maps are used to build the risk prediction pipeline and predict the binary risk cohort label for drivers in the held-out set (approximately 250 drivers). For the drivers in each cohort, the average cohort risk was represented in terms of the average number of traffic citations or accident records in the past five years. Results are shown in Tables 5 and 6 (for the proposed framework) and Fig. 4 (for the baseline method).

The main takeaways are summarized as follows.

- **Best combination of feature groups:** The results imply that the features maps T1 to T14 (i.e., the feature groups 1, 2, and 3) provided the best risk cohort classification results, considering the “difference between average risk scores” obtained for low- versus high-risk cohorts. It is also interesting to note that about 76% of drivers in the low-risk cohort do not have any traffic records (i.e., at-fault traffic accidents and citations) in the past five years according to Tables 5 and 6.
- **Best single feature group:** The fifth feature group that includes feature maps T18, T19, and T20 provided the best risk cohort prediction results (based on difference in terms of average risk score for low-risk versus high-risk cohort). This group of feature

maps are the ones based on speed and angle change, but for turn segments. This is an interesting observation, because *turns* represent an important aspect of “driving behavior”, and they seem to provide a very strong signal to identify risky drivers. In addition, around 71% of low-risk drivers have no traffic records in the past 5 years, while the percentage for high-risk drivers is about 49% (CNN model, Table 6). We also achieve improved cohort balance compared to the best feature group combination discussed earlier. Achieving balanced cohorts is a significant objective in our proposed framework, successfully attained through the design, e.g., using a right threshold for “labeling confidence”.

- **Impact of deviation maps and risk re-labeling:** The results presented in Fig. 4 underscore the crucial role played by two key components of our approach: (1) deviation feature maps, and (2) risk re-labeling. Notably, none of the combinations of feature maps yielded superior results to that of having two highly imbalanced cohorts, where the average risk score for the larger cohort closely aligns with the overall average risk score in the held-out set (indicated by the blue dashed line). Furthermore, these findings emphasize that relying solely on raw telematics data and original (i.e., weak or partial) risk labels may not suffice for reasonable driving risk prediction. Instead, appropriate transformations are essential to ensure the success of the prediction task, akin to the strategies proposed in this paper.
- **Impact of contextual information: Road type:** Comparing feature groups G1 and G2, it is observed that including road type information as a contextual factor improves the separation between low- and high-risk drivers based on average risk score and the proportion of drivers with no records in each cohort. This highlights the importance of contextual information in determining driving risk.
- **Impact of contextual information: Daylight:** Comparing feature groups G1 and G3 reveals that adding daylight information (day-time versus night time) has a significant impact on determining driving risk. The inclusion of this information changes the classification of some drivers from low-risk to high-risk, leading to an increase in the proportion of drivers with no past records and a decrease in average risk for the low-risk cohort.
- **Impact of contextual information: Road shape:** Road shape encoded as turn maneuvers improved the distribution of drivers between low-risk and high-risk cohorts when represented by speed and angle change (G5), while other combinations (G4 and G6) resulted in worse outcomes compared to G1. This highlights the importance of the choice of data attributes in representing telematics data.

Table 5

Risk cohort prediction results for the held-out set using the GBC model. The table displays two risk cohorts: *low-risk* and *high-risk*. The *No Records %* column shows the proportion of drivers with no traffic accident or citation records in each cohort. The *Average Risk* represents the average number of (citation and at-fault accident) records per driver in each cohort. The *P-Value* indicates the significance of the difference between average risk scores obtained for the low- and high-risk cohorts. Rows colored in light green represent the cohorts that provided the best results according to our problem formulation in Section 3.

Feature groups	Low-risk drivers %	High-risk drivers %	Low-risk no records %	High-risk no records %	Avg. risk of low-risk	Avg. risk of high-risk	P-value
G1	43%	57%	64.95%	55.12%	0.526	0.764	0.000
G2	38%	62%	63.79%	51.06%	0.569	0.819	0.000
G3	33%	67%	67.12%	54.48%	0.452	0.779	0.000
G4	41%	59%	58.24%	60.15%	0.659	0.662	0.214
G5	51%	49%	67.83%	50.46%	0.443	0.890	0.000
G6	43%	57%	64.95%	55.12%	0.557	0.740	0.000
G1G2	32%	68%	65.31%	51.46%	0.449	0.854	0.000
G1G3	41%	59%	63.33%	55.47%	0.522	0.773	0.000
G1G4	48%	52%	61.11%	57.76%	0.556	0.759	0.000
G1G5	48%	52%	65.74%	53.45%	0.500	0.810	0.000
G1G2G3	25%	75%	75.68%	48.65%	0.297	0.874	0.000
G1G2G4	32%	68%	68.75%	50.00%	0.500	0.827	0.000
G1G3G4	43%	57%	65.96%	53.23%	0.489	0.806	0.000
G1G2G3G4	24%	76%	75.00%	49.11%	0.306	0.866	0.000
All	26%	74%	73.68%	49.09%	0.316	0.873	0.000

Table 6

Risk cohort prediction results for held-out set using CNN model.

Feature groups	Low-risk drivers %	High-risk drivers %	Low-risk no records %	High-risk no records %	Avg. risk of low-risk	Avg. risk of high-risk	P-value
G1	70%	30%	63.46%	50.00%	0.551	0.912	0.000
G2	36%	64%	70.37%	47.96%	0.426	0.888	0.000
G3	42%	58%	67.03%	52.76%	0.473	0.811	0.000
G4	74%	26%	61.45%	53.45%	0.596	0.845	0.000
G5	48%	52%	71.03%	48.72%	0.364	0.932	0.000
G6	44%	56%	61.62%	57.60%	0.606	0.704	0.001
G1G2	39%	61%	65.00%	50.00%	0.483	0.880	0.000
G1G3	48%	52%	66.67%	51.33%	0.476	0.850	0.000
G1G4	49%	51%	64.55%	54.39%	0.509	0.807	0.000
G1G5	57%	43%	66.41%	50.00%	0.477	0.906	0.000
G1G2G3	28%	72%	76.19%	47.17%	0.286	0.906	0.000
G1G2G4	37%	63%	67.86%	48.96%	0.464	0.875	0.000
G1G3G4	27%	73%	69.49%	54.72%	0.390	0.774	0.000
G1G2G3G4	28%	72%	71.43%	49.06%	0.381	0.868	0.000
All	28%	72%	73.17%	48.60%	0.317	0.888	0.000

- **The better risk classifier:** According to the results, the use of the CNN model resulted in better risk prediction outcomes. Based on feature group G5, the two classifiers found relatively similar-sized low- and high-risk cohorts (see the first highlighted row in Tables 5 and 6). However, a larger gap is observed between the cohorts found by the CNN model when comparing their average risk and the percentage of drivers with no records (see Figs. 5-a and 5-c). This pattern persists when analyzing the results based on a combination of feature groups G1, G2, and G3 (see Figs. 5-b and 5-d). The superior results of the CNN model are assumed to be due to its ability to better capture and encode spatio-temporal patterns when presented with a tensor-shaped input.
- **Significance of difference in average risk scores:** To assess the significance of the difference between average risk scores obtained for low- and high-risk cohorts, an independent samples t-test is conducted for each case. The significance level was set at 99%, and the results presented in Tables 5 and 6 (the *p*-value column) demonstrate a significant difference in average risk scores between the low- and high-risk cohorts in all cases except one (i.e., for feature group G4 and the GBC model). Once again, these findings underscore the effectiveness of the proposed framework in identifying cohorts that exhibit meaningful distinctions.

It is worth noting that in most cases, the proportion of drivers with no crashes or traffic citations was higher in the low-risk group than the high-risk group. However, some low-risk drivers had records and vice versa, indicating that having or not having records does not determine risk level. Nevertheless, most drivers with records may be considered high-risk.

6.3. Implications for practice

In this section, practical applications of the proposed framework in real-world scenarios are explored. Two examples are presented to illustrate how the framework can be effectively utilized.

Scenario I: Personalized Insurance Premium Adjustment One compelling application of the proposed framework is in the auto-insurance industry to adjust premium costs based on individual driving styles. Consider an insurance company with a vast customer base, each owning one or more vehicles, and the ability to collect telemetry data to monitor their driving behavior over a certain period. Once the company gathers enough telemetry data, it can implement our proposed framework by creating a reference and modeling set, and subsequently constructing a risk cohort classifier, as detailed in this paper. For any new customer, the company can easily monitor their driving data for a brief period to determine the appropriate risk cohort the driver belongs to. This inferred risk cohort label can then serve as a new, and potentially strong, signal to be factored into the premium pricing, complementing traditional factors such as demographics and past records. As discussed earlier in the paper, our framework allows for customization, including the selection of attributes to collect during driving and the number of risk cohorts. These customizations can be tailored to specific areas, customer behavior, and the unique needs of the company, enhancing its ability to offer personalized insurance premiums and optimize risk assessment.

Scenario II: Driving-based Feedback to Enhance Driving Skills In a similar vein to Scenario I, let us consider an insurance company with a large user base and the capacity to gather telemetry data on a

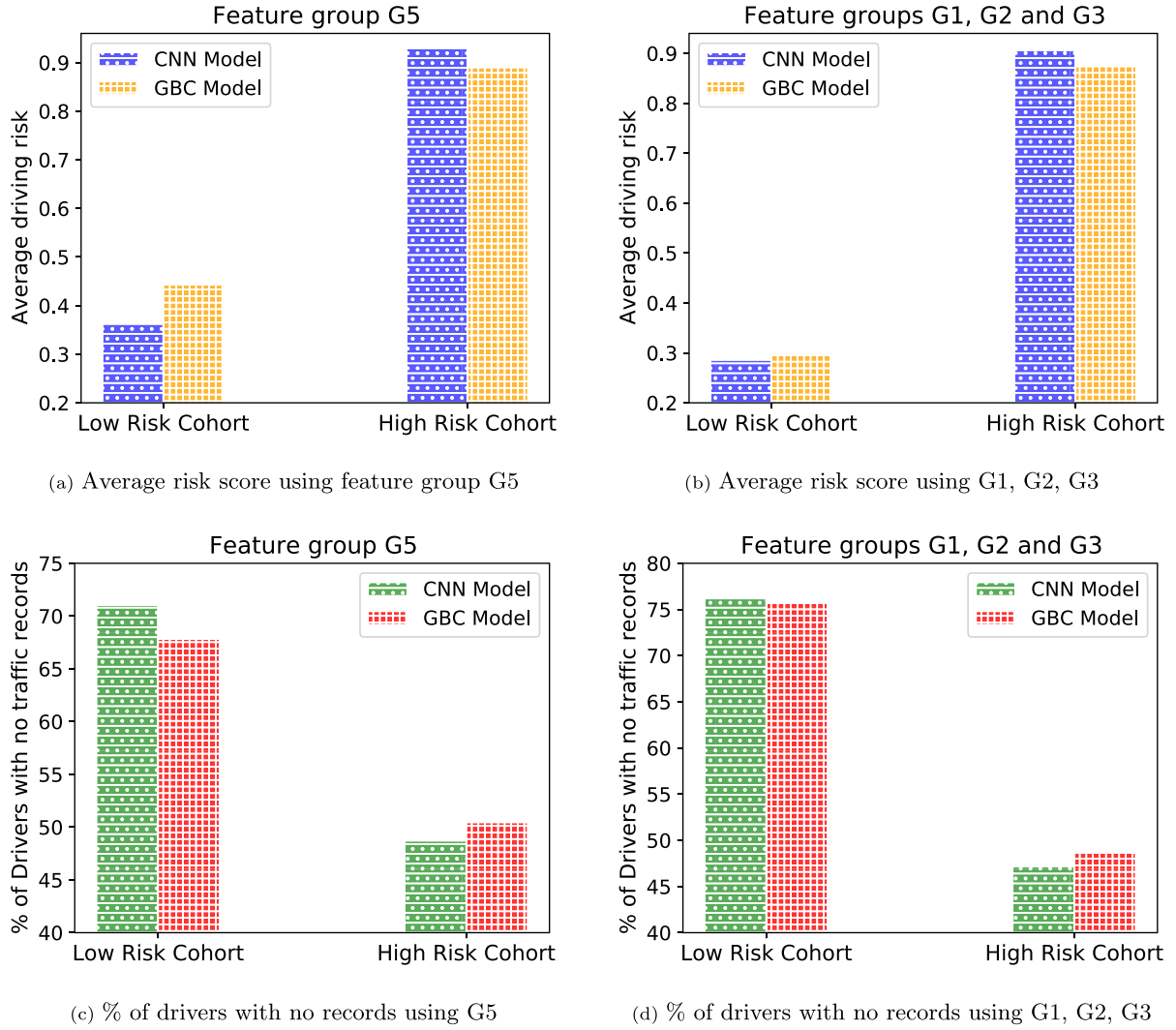


Fig. 5. Comparing GBC and CNN models on average risk scores in low- and high-risk cohorts (a and b), and percentage of drivers with no citation or (at-fault) accident records in each cohort (c and d). The CNN model outperforms GBC, finding better cohorts with the two best feature groups. Average risk score represents at-fault traffic accidents and citations per driver over five years, including the data collection period. Results are based on drivers in the held-out set (visit Section 5.1 for details).

significant scale, enabling the implementation of the proposed framework described in this study. When a new driver chooses to participate in the company's program, their driving behavior is monitored for a period to determine a personalized insurance premium. Throughout the data collection and risk cohort assignment process (which may occur multiple times via a feedback mechanism), the driver gains insights into how their driving behavior contributes to their assignment to a specific risk group. While not addressed in this study, it is worth noting that the framework can be adapted to incorporate explainability. This enhancement would elucidate the driving patterns and contexts that influence the classification into a particular risk cohort. For instance, a driver might discover, through interaction with the proposed framework, that their driving behavior is perceived as safe on low-speed roads but flagged as risky on high-speed roads. Such valuable insights can empower a driver to improve their skills and behavior progressively, ultimately transforming into a safer driver with a reduced likelihood of being involved in at-fault crash. By leveraging the feedback provided by the framework, drivers can fine-tune their driving habits over time, leading to safer road practices and potentially contributing to a reduction in overall insurance risk.

7. Summary and future work

In this paper, we introduced a novel driving risk prediction framework that utilizes telematics and contextual data. The framework comprises a modeling and prediction process. Key steps include constructing contextualized views of trajectories, generating deviation-based views for drivers, refining weak risk labels using a data-driven approach, and building a risk cohort classifier. The prediction process employs the trained risk cohort classifier to predict driving risk cohorts using contextualized telematics data. An important aspect is the risk label refinement step, which uses a novel data-driven approach to improve the quality of the risk labels.

Regarding contextual data, the framework uses road type, road shape, and daylight information. Incorporating additional data like weather and traffic could enhance the framework by creating more feature and deviation maps. The framework is flexible, allowing any contextualized view of a trajectory to be used as input and permitting adjustments in the number of risk cohorts for different regions. Furthermore, the clustering algorithm, classification method, and deviation calculation formulation can be customized based on specific needs.

While the telematics data used in this research is primarily derived from the CAN-bus and GPS sensor, other sensors like accelerometer can

offer valuable insights into driving behavior. High-quality accelerometer data collected at high frequency rates can capture interesting patterns, such as lane changes and sudden jerky movements, which, when contextualized, can improve driving behavior analysis and risk prediction. Additionally, addressing the challenge of obtaining fine-grained risk labels for individual trajectories or sub-trajectories could lead to more accurate and useful risk models. Having risk data available for each trajectory would benefit real-time applications and help prevent dangerous driving actions leading to crashes.

Considering future directions, we recognize that the current framework assumes each driver owns only one vehicle and that no other drivers share that vehicle under an insurance policy. Acknowledging that this simplification may not directly apply in real-world scenarios where vehicle sharing occurs, the proposed framework can be extended to accommodate such situations. One promising approach involves segmenting trajectories by drivers using additional per-trajectory data, thus enabling the adaptation of the framework for improved modeling and prediction under these circumstances. By addressing this limitation, our framework will be even more robust and applicable to a wider range of real-world driving scenarios.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: This work is supported by a grant from the Nationwide Mutual Insurance (GRT00053368), and another from the Ohio Supercomputer Center (PAS0536).

Data availability

The data is planned to be published on Kaggle under <https://www.kaggle.com/sobhanmoosavi>.

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